

Farmers' knowledge, perceptions and management of chili pepper anthracnose disease in Bangladesh

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ABSTRACT

Plant diseases cause major crop yield losses and affect food security in lower-income countries. A limited understanding by farmers about the identity of plant diseases, transmission pathways and appropriate management methods is a key, but underexposed, problem. This study uses a sample of 260 farmers producing chili pepper in Bangladesh to elucidate their knowledge, perceptions and management methods regarding chili anthracnose—a fungal disease caused by *Colletotrichum* species. The results show that virtually all farmers had experienced typical symptoms of anthracnose disease and farmers reported average crop yield losses of 4%. Yet, only 22% of the farmers knew that this was a fungal disease and only 25% could tell how the disease spreads in the field. Farmers solely relied on chemical pesticides to control anthracnose, while the use of biopesticides, soil covers and crop rotations was negligible and only 7% of the farmers used labelled seed. A sound understanding of farmers of the crop diseases they face, their transmission pathways, and appropriate management methods is essential for establishing effective pest control. These results clearly show much room for improving chili productivity and the income and health of farmers, by building their capacity to deal with anthracnose disease.

1. Introduction

Crop pests and diseases are a key factor affecting global food security (Strange and Scott, 2005; Savary et al., 2012). It is estimated that about 26–40% of the world's crop production is lost because of weeds, pests and diseases despite the crop protection measures applied by farmers (Oerke, 2006). Farmers in lower-income countries are particularly affected by this. Not only do they have a more limited choice of available management options, but pest and disease pressure are also generally higher in tropical and subtropical areas of the world, which is a problem that is likely to worsen as a result of climate change (Deutsch et al., 2018).

Another key factor affecting many farmers in lower-income countries is their lack of knowledge of the pests and diseases they are dealing with, including the mechanisms by which these spread and effective management methods. For instance, Uwamahoro et al. (2018) studied farmers' knowledge of potato bacterial wilt (PBW) in Rwanda, a disease caused by *Ralstonia solanacearum*. They showed that potato farmers

were generally able to recognize the disease, but less than half of the respondents knew that it is spread through soil, tubers, seed and the use of infected farm tools. Echodu et al. (2019) showed that farmers growing sweet potatoes in four East African countries were unable to correctly associate disease symptoms such as leaf wilting, curling and yellowing with viral, bacterial or fungal diseases. Furthermore, Schreinemachers et al. (2015) studied farmers' knowledge and perception of plant viruses in vegetable crops in India, Thailand and Vietnam and found that very few farmers were able to associate typical disease symptoms with plant viruses and even fewer knew the role of whiteflies (*Aleyrodidae*) and thrips (*Thysanoptera*) as vectors and that these needed to be controlled to contain the disease in the field.

These examples show that farmers' lack of correct knowledge about pests and diseases poses serious limitations to crop performance. Consequently, farmers may apply management methods that are ineffective or even aggravate the problem. It is therefore essential to develop a sound understanding of farmers' knowledge of pests and diseases and the management methods they employ to inform the development and

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dissemination of appropriate management strategies.

This study contributes to the existing literature by documenting the knowledge, perceptions and practices of smallholder farmers in Bangladesh producing chili peppers. The study focuses on anthracnose, which is the most important disease affecting chili pepper production globally (Rahman et al., 2011; Suwor et al., 2015; Ali et al., 2016). The causal agent of the disease is the fungus *Colletotrichum* spp., which naturally inhabits the soil. Plants usually get infected when infected soil touches the stems, leaves or fruits through the use of unclean farm tools, hands, or water splashing (Saxena et al., 2016). Anthracnose disease also affects a wide range of other fruit and vegetable crops, causing serious economic damage, and the results of this study therefore have wider relevance.

Typical anthracnose symptoms on chili fruits are sunken necrotic tissues with concentric rings, which turn brown and then black due to the formation of setae and sclerotia (Agris, 2005; Suwor et al., 2015). The infection can predispose the fruit to secondary infections by other fungal pathogens, including *Aspergillus flavus*, thereby creating a possibility for the production of aflatoxins, which are a serious hazard to human health (Roy et al., 2013; Rashid et al., 2015). Latent infection can also occur starting from flowering to fruiting stages and symptoms are particularly severe under cool and wet weather conditions as experienced during the monsoon season.

The use of mulches to reduce water splashing, crop rotations with non-host species, and the use of clean farm tools and farm worker hygiene can effectively reduce anthracnose damage in chili production (Roberts et al., 2001; Nelson, 2008; Than et al., 2008a,b; Choudhary et al., 2013; Pranshanth et al., 2014; Ali et al., 2016). Fungicide sprays can be effective but are often washed away by unpredictable rains. However, the proper use of any of these methods requires a basic understanding on the side of the farmer about the disease, which is what this study elucidates.

2. Materials and methods

2.1. Conceptual framework

This study applied the concepts of knowledge, perceptions and practices, which provides a structured framework to understand the interaction between human behavior and an environmental challenge (Ekman and Walker, 2008). Such assessment informs what people know about the problem, how they perceive its effects, and what they currently do about it. The method has been widely applied to study farmers' management of crop pests and diseases (for instance, Adam et al., 2015; Mendesil et al., 2016; Midega et al., 2016).

Knowledge refers to farmers' understanding of the disease, its causal agent, method of spread and conditions conducive to its spread, and management practices. Perceptions (or attitudes) here refer to farmers' beliefs regarding pesticide safety, the spread of the plant disease, and possible misconceptions stemming from incorrect knowledge (for instance, a belief that anthracnose is mainly spread by the wind). Practices refer to management decisions that demonstrates knowledge and perceptions. The framework assumes that a change in farmers' practices is the cumulative result of a change in farmers' knowledge and attitudes. The information obtained can help to tailor interventions to close certain knowledge gaps or to correct prevailing misconceptions.

2.2. Sample selection

Chili pepper is an important crop in Bangladesh, grown on about 100,000 ha with a total production of 136,872 tons annually (BBS, 2017). The crop is grown during the winter season from mid-November to mid-January and during the rainy season from mid-April to mid-July (BBS, 2017). Bogra District was selected for this study because it is one of the major chili pepper producing areas of the country. Four out of 12 sub-districts were purposively selected for this study (Fig. 1) in

consultation with the Department of Agricultural Extension and the Spice Research Center of the Bangladesh Agricultural Research Institute, which is also located in Bogra. From these, 13 unions and 27 villages were selected to capture the main chili producing areas of the district. A random sample was then taken of 10 farmers per village using a list of all chili pepper farmers in a village. The total sample included 260 farmers and can be considered as representative for chili pepper production in Bogra District and indicative for chili pepper production in Bangladesh.

2.3. Data collection and analysis

Data were collected from January to February 2019 using a structured questionnaire developed in English and translated into Bangla. The questionnaire was first tested with 12 chili pepper farmers selected from outside the study area and subsequently refined. The questionnaire was administered by a team of trained and experienced enumerators. Interviews were conducted with the person in the household who usually made the decisions regarding input use, pest control, and selling of chili.

Before the start of the interview, respondents were informed about the purpose of the study, confidentiality of the data, approximate length of the interview, and contact information of the research team. Respondents were explained that their participation was voluntary and they could stop the interview at any time without consequences. Respondents were asked to approve of the process by signing a consent form before the start of the interview. Participation in the survey was virtually risk-free.

The questionnaire recorded data on individual and household socio-economic characteristics, agronomic practices in chili pepper production, inputs used and output sold, methods used for pest management, and perceived yield losses from pests and diseases. The study also asked detailed questions about the use of chemical pesticides such as the products applied, spraying frequency, their perception and understanding the possible health risks, their use of protective gear during spraying and possible symptoms of poisoning experienced after spraying pesticides (such as headache, stomach cramps, blurred vision, muscle twitching). Questions on chili pepper production referred to the most recent rainy (monsoon) season (called the *khari*) from mid-April to mid-July 2018.

Pests and diseases in chili pepper were identified using a photo tool. The study used 21 color photos showing disease symptoms and insect pests and the damage they typically cause in chili pepper. For each photo respondents were asked whether it caused regular damage to their chili production. To quantify the perceived losses from a pest or disease, respondents had to rank the selected photos in descending order of the damage caused (following Schreinemachers et al., 2015). The respondents were then asked to estimate the total damage to crop yields by all pests and diseases combined. Next, they were given twenty coins, representing the total damage they just estimated, and they were asked to allocate these to the photographs. These data were used to estimate the perceived yield losses from each pest and disease. Farmers' pest and disease identification and damage quantification might be quite different and less accurate than that of a trained expert. It provides nevertheless important information, because farmers decide on a course of action based on what they think the problem is, not on the actual problem.

Respondents were also shown photos of various anthracnose disease symptoms and were asked if they had ever experienced these issues in their field, how often they experienced it in the last five chili production cycles, what they thought were the causes of these symptoms, and how they tried to manage it if a field was affected.

Interviews took between one and two hours, depending on the scale of production and number of pesticides applied. Data from the questionnaire were encoded, entered in a customized spreadsheet and checked for outliers and consistency. Data were analyzed by calculating means, standard deviations, frequencies and percentages.

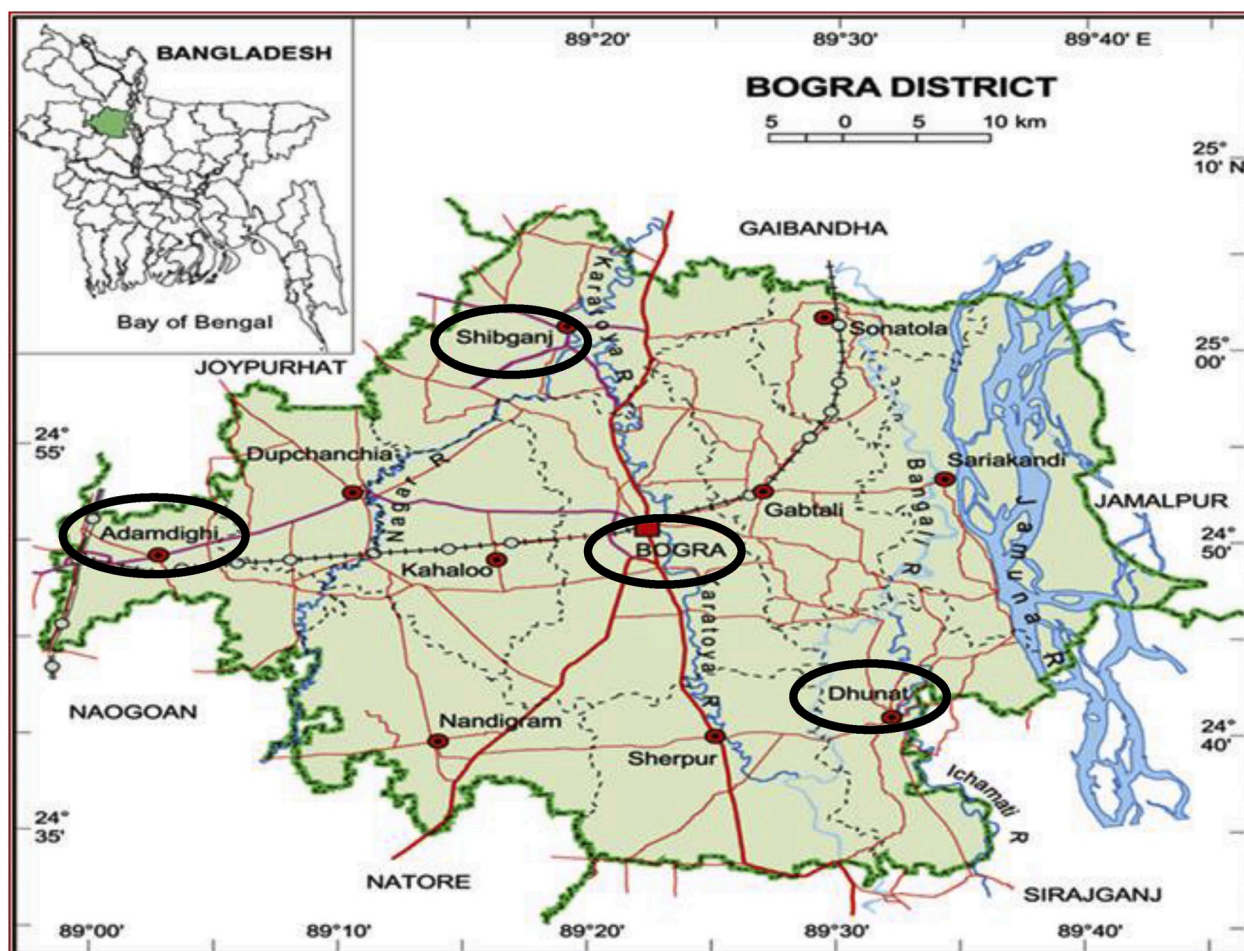


Fig. 1. Map of Bogra District, Bangladesh showing the four sub-districts included in the study (Khatun et al., 2018).

3. Results

3.1. Sample characteristics and chili pepper production

All 260 chili pepper farmers were men, averaging 43 years of age and an average of 5 years of schooling (Table 1). Most farmers (62%) were able to read or write. The average households had 6 persons, which is above the national average of 4.1 (BBS, 2017). For 57% of the households, at least one person was a member of an organization and 48% had

Table 1

Characteristics of the sample of chili pepper farmers in Bogra, Bangladesh, 2018, in the percentage of farmers unless indicated otherwise (n = 260).

Household characteristic	Mean	SD
Age (years)	43.5	11.9
Years of education	5.0	4.3
Able to read or write	62	
Household size (persons)	5.5	1.7
Dependency ratio	52.6	49.2
Total cultivated land area (ha)	0.62	0.48
Cultivated land area owned (ha)	0.47	0.53
Household is a member of an organization	57	
Currently has a loan	48	
Main occupation in agriculture	69	
Main occupation outside agriculture	31	
Years of residence in the current village	43.1	12.2
Years of commercial vegetables production	21.8	11.1
Years of commercial chili production	13.1	7.2

Notes: Dependency ratio defined as $100 \times (\text{persons aged } 0-16 + \text{persons aged } \geq 60) / (\text{persons aged } 17-59)$. SD=Standard deviation.

taken a loan. Agriculture was the main occupation for 69% of the respondents. The average cultivated land area was 0.62 ha of which 0.47 ha was own land, indicating that the sample represents smallholder farmers. The average farmer in the sample had 22 years of experience in commercial vegetable production, of which 13 years were in chili production.

The average farmer in the sample produced chili pepper on 0.06 ha of land during the *kharif* season and for a period of 6.5 months from planting to the last harvest (Table 2). Most farmers (89%) used seed rather than seedlings to raise a new crop. For those that used seed, 74% used own seed saved from the previous harvest while the rest purchased seed, of which 93% was not labelled and 7% was labelled seed. Therefore, only 17 farmers out of a sample of 260 farmers (7%), bought seed from the formal seed sector.

On average, chili pepper was grown 2.7 times on the same land during the past 3 cropping seasons, which points at a lack of rotation with other crops. About 4% of the farmers had received training in chili production, but none had received training in integrated pest management methods. The intercropping of chili pepper with other plants was uncommon (8%). The average producer harvested chilies about 16 times per cropping cycle. Income from selling chili pepper accounted for about 17% of the annual household income (Table 2).

Hired labor represented the highest single variable cost in chili pepper production, accounting for 15% of total cost (Table 3). This was followed by the cost of chemical pesticides (12%), mineral fertilizers (8%) and organic fertilizers (7%). No farmer used biopesticides. Fixed costs included the opportunity cost of family labor and land, which jointly accounted for 46% of the total production costs. The average yield of chili pepper was 19 t/ha. The average seasonal gross margin and

Table 2

Characteristics of chili pepper production in Bogra District, Bangladesh, 2018, in % of all farmers in the sample unless indicated otherwise.

Items	Mean	SD
Planted area (ha)	0.06	0.03
Length of growing period (months)	6.5	0.8
Type of planting material (% of all farmers) (N = 260):		
– Seedlings	11	
– Seed	89	
Source of seed (% of farmers who used seed) (N = 231):		
– Own saved seed	74	
– Purchased seed	26	
Type of seed and seedlings (N = 260):		
– Labelled (formal seed source)	7	
– Non-labelled (informal seed source)	93	
Times the crop was grown in 3 last cropping seasons	2.7	1.0
Received training in production	4	
Received training in integrated pest management	0	
Intercropped chili with other plants	8	
Harvests per season	15.7	3.7
Income from chili pepper (% of annual cash income) ^a	16.7	9.1

Notes: ^a This is a rough estimate because the survey did not do a detailed quantification of all income sources. Based on farmers' perception of its contribution, not from a quantification of all income sources. SD=Standard deviation.

Table 3

Cost and returns to chili pepper production during the *kharif* season in Bogra, Bangladesh, 2018, in US dollars/ha.

Items	Mean	SD	% of total cost
Costs:			
A. Variable cost:	2388	959	53
– Land preparation	83	48	2
– Seed/seedlings/nursery supplies	156	253	3
– Organic fertilizers	321	205	7
– Mineral fertilizers	356	174	8
– Irrigation fees	66	46	1
– Plant hormone	13	28	<1
– Chemical pesticides	525	351	12
– Bio-pesticides	0	0	0
– Other variable costs	173	180	4
– Hired labor	694	510	16
B. Fixed cost:	2080	737	47
– Opportunity cost of family labor*	1483	731	33
– Opportunity cost of land*	598	167	13
C. Total cost (A + B)	4468	1313	100
Returns:			
Total harvested yield (t/ha)	18.75	8.50	
D. Output value	8742	4931	
Profitability:			
G. Gross margin (D-A)	6354	4483	
H. Net return (D-A-B)	4274	4335	
I. Benefit-cost ratio (BCR) (D/C)	1.96	0.86	
J. % of farmers with a positive net return	86		

Notes: Bangladeshi taka (BDT) was converted to US dollar (USD) using the exchange rate of 83.75 BDT/USD (August 2018). SD=Standard deviation. * The opportunity cost of family labor was calculated based on the wage rate family members could obtain if working outside the own farm. The opportunity cost of own land was calculated based on the rent a farmer could receive.

net return were estimated to be USD 6354 per ha and USD 4274 per ha, respectively. The benefit-cost ratio indicates that the average farmer earned 1.96 USD for every 1 USD invested in chili pepper production. Overall, 86% of the farmers earned a profit from chili pepper production.

3.2. Pest incidences and losses

Farmers reported an average of 11 different insect pests and diseases in chili pepper during the most recent production season, as identified using photos. Combined, these pests and diseases reduced farmers' yields

by about 23%, which translates into USD 2663 per ha or USD 160 per household. These are farmers' perceived yield losses, which may be different from actual yield losses. The results confirm that anthracnose is the most important pest or disease affecting chili production in Bangladesh. Virtually all farmers (99%) had encountered problems with anthracnose in their field. The disease accounted for 18% of the perceived total yield losses (of 23%). In other words, it reduced chili pepper yield by 4.1%, which is an economic loss of USD 479 per ha. Other major pests included mites, army cutworm, boll worm/fruit borer/pod borer (see Fig. 2).

3.3. Farmers' knowledge and perceptions about chili anthracnose

Of the farmers, 99% said that they had observed anthracnose disease symptoms in their chili production during the last five cropping cycles (Table 4). On average, farmers had observed these symptoms 3.7 times during these five cycles. Asked whether they perceived the severity of these symptoms to have changed over the past five years, 43% indicated no change, 33% an increase, and 24% a decrease in severity.

About half of the respondents (56%) was unable to name any cause of the anthracnose disease symptoms, 17% said weather conditions cause the problem, and 24% could correctly explain that a fungus causes it. Only 22% of the respondents was able to explain what a fungus is.

Furthermore, asked how the symptoms shown on the photo spread in the field, only about a quarter of the respondents were able to describe correctly how the disease spreads. Although wind and insects can potentially spread the *Colletotrichum* pathogens, the most common pathways are water splashing (either from irrigation or rain) and infected farm tools. Yet, only 2% mentioned farm tools as spreading the disease (Table 4).

3.4. Chili pepper anthracnose management practices

The application of chemical pesticides was the most common method to control anthracnose diseases in chili production, used by all farmers (Table 5). Only 1% of the respondents indicated to have used cultural control methods such as mulches to cover the soil, resistant varieties or quality seed to prevent or control anthracnose. Apart from using pesticides, 9% of the farmers harvested chilis at the immature stage before the disease symptoms become more apparent (see Table 5).

When farmers observe anthracnose infection in their chili pepper field, 54% said that they would still harvest their field as usual, but sort after harvest and discard the infected fruit. Another 45% indicated that they would leave the infected fruit in the field and 1% (2 farmers) said they would harvest chilis at the immature (green) stage instead of the mature (red) stage. This is understandable because in Bangladesh and other countries, the market demand and price for red chili fruits is higher and more constant than for green fruits. One farmer said that infected fruit could be used to produce chili powder while two farmers said they would still sell the infected fruit, which are practices that could be hazardous to consumers. Farmers were aware that there is a health risk from eating chili fruits infected with anthracnose as 97% of the respondents said that it is unhealthy. Most respondents were unable to explain why contaminated fruits are harmful, but 15% was knowledgeable about aflatoxins while 23% thought that infected chili fruits are unhealthy because of pesticide spraying, which could also be true (Table 5).

The average application rate of synthetic pesticides was 21.6 kg of formulated product per hectare per production cycle. The average farmer sprayed his field 14 times. Farmers were moderately satisfied with the level of control this achieved with 10%, indicating that the level of control was 100% and 28% saying that it gave more than 80% control. Yet, 82% of the respondents reported that the effectiveness of chemical pesticides against anthracnose disease has reduced over time (Table 6).

In terms of worker safety, chili producers indicated that they would typically cover their upper body with a long-sleeved shirt (89%), while

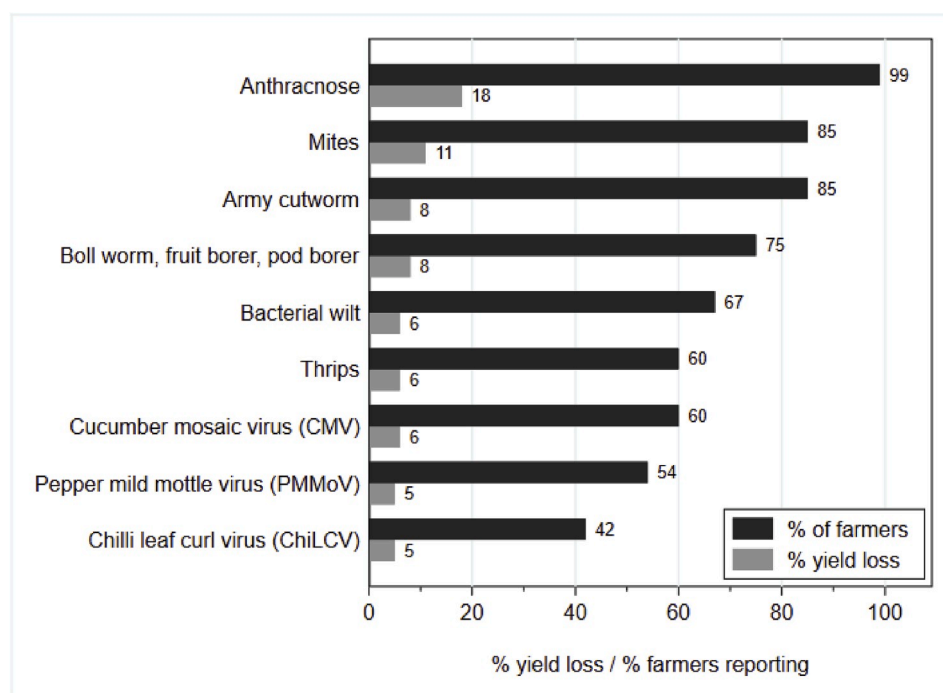


Fig. 2. Major pests and diseases in chili pepper production as perceived by farmers in Bogra District, Bangladesh during the 2018 *kharif* season ($n = 260$).
Notes: Yield losses are based on farmers' perception and are expressed as a percentage of the total yield losses from all pests and diseases, estimated to be 23%. Percentages do not add up to 100% because it only shows a selection of the major pests and diseases.

Table 4

Farmers knowledge and perceptions regarding chili anthracnose, in % of farmers unless indicated otherwise.

Type of knowledge or perception	% of farmers
Observed anthracnose symptoms during last 5 crop cycles	99
Number of times it was observed during last 5 cropping cycles	3.7
Has the disease severity changed in the past 5 years?	
– Increased	33
– No change	43
– Decreased	24
Perceived cause of the disease:	
– Don't know	56
– Weather/humid weather	17
– Fungus	24
– Insect	2
– Soil	2
Farmer is (more or less) able to explain what a fungus is	22
Farmers' perception about how the disease spreads:	
– Don't know	49
– Spreads by wind	35
– Spreads by rain	25
– Spreads by insects	16
– Spreads by farm tools	2
– Spreads by water in the field	2

52% wore a long pair of trousers, and 29% used a mouth and nose cap/mask. However, head, eyes, hands and feet tended to be exposed during spraying (Table 6).

4. Discussion

Our survey results for 260 chili pepper growers in Bangladesh show that chili pepper producers have very limited knowledge of anthracnose. Virtually all farmers have experienced anthracnose disease symptoms in their field, but only a quarter could correctly associate these symptoms with the disease. Furthermore, only about a quarter of the sampled farmers knew that it is caused by a fungus and only about a quarter had some knowledge about the transmission pathway—associating it with

Table 5

Anthracnose management practices of chili pepper producers in Bogra, Bangladesh, 2018, in % of farmers ($n = 260$).

Management method	% of farmers
Methods used to prevent or control chili anthracnose:	
– Synthetic pesticides	100
– Biopesticides	0
– Crop rotation	0
– Avoid splashing water	0
– Avoid sprinkler irrigation	0
– Avoid soil contact with the fruits	1
– Select a resistant variety	1
– Use quality seed	1
– Use nursery to raise seedlings	0
– Use soil cover/mulch	1
– Harvest chilis at green stage	9
What to do with anthracnose infected chili fruits?	
– Harvest at immature/green stage	1
– Do not harvest, leave in the field	45
– Harvest normally but discard after sorting	53
– Harvest normally and mix with unaffected fruits	1
Are infected fruits consumed within your household? (yes)	3
Reasons for not consuming the affected chili fruit ((% of farmers who did not consume the infected fruit) ($n = 253$)):	
– The taste is not good	13
– It does not look nice	27
– It is unhealthy because of pesticide	23
– It is unhealthy because of aflatoxin	15
– It is unhealthy but cannot explain why	62
– Others	1

rain, but not necessarily with the correct explanation that rain splashes *Colletotrichum*-infected soil on the plant, which then causes anthracnose disease. In particular, only 2% of the respondents were aware that infected farm tools can spread the disease. These findings are in agreement with those of other similar studies that highlighted the limited state of knowledge of farmers in developing countries about plant diseases (Abang et al., 2014; Adam et al., 2015; Schreinemachers et al., 2015).

Table 6

Pesticide use practices among chili pepper producers in Bogra, Bangladesh, 2018, in % of farmers unless indicated otherwise (n = 260).

Items	Mean	SD
Quantity of pesticides used (kg/ha) ^a	21.6	10.1
Pesticide sprays per production cycle	14.3	5.6
Perceived effectiveness of pesticides against anthracnose:		
– Useless (0% control)	<1	
– Slightly useful (20%)	9	
– Moderate useful (21–80%)	52	
– Useful (>81%)	28	
– Very useful (100%)	10	
Observed a reduction in the effectiveness against anthracnose	82	
Use of protective gear during pesticides spraying:		
– Long-sleeved shirt	89	
– Long trousers	52	
– Mouth and nose cap/mask	29	
– Raincoat or coverall	10	
– Hat	3	
– Eye cover	3	
– Gloves	1	
– Respirator	1	
– Rubber boots	0	

Note: ^a Includes herbicides, insecticide and fungicides.

Our finding that anthracnose is the most serious pest and disease problem affecting farmers in Bangladesh is consistent with evidence from literature which shows that this disease is one of the most economically important diseases accounting for 10–80% of the yield losses from pests and diseases in chili production in developing countries (Than et al., 2008a,b; Ali et al., 2016). Another study reported that anthracnose accounted on average for 30, 25 and 10% of yield losses in chili production in Thailand Vietnam and India, respectively (Schreinemachers et al., 2015).

Farmers in Bangladesh have very limited access to extension services, and where present, extension services do not prioritize chili pepper. Our study also showed that chili pepper is continuously grown on already infested soils, which is likely to increase the incidence and severity of anthracnose disease and also of other soil borne diseases. The estimated amount of pesticide use, 21.6 kg per ha per year, is high and the use of protective gear during spraying is very limited, which confirms the findings of previous studies for Bangladesh that showed unsafe pesticide spraying practices (Dasgupta et al., 2007; Schreinemachers et al., 2016; Akter et al., 2018).

On the positive side, the lack of farmer knowledge, lack of application of safe pest management practices and formal seed and seedling distribution system provide relatively straightforward opportunities for improvements in chili pepper production. Knowledge about anthracnose is readily available and can be disseminated to farmers through appropriate channels (such as extension systems) while taking into account that only about 62% of the chili pepper farmers can read. Previous studies have shown that crop rotation, use of biopesticides, and quality seed are effective methods to control anthracnose (Roberts et al., 2001; Nelson, 2008; Than et al., 2008a,b; Choudhary et al., 2013; Pranshanth et al., 2014). Wider application of mulches to cover the soil and prevent water splashing and especially the use of crop rotations can provide quick gains in disease prevention and control. Our study found that only 7% of the farmers purchased seed from the formal sector, which shows large opportunities for the emerging vegetable seed sector in Bangladesh to supply farmers with quality seed of improved varieties and hybrids, especially those which are tolerant/resistant to chili anthracnose.

Chili pepper is an economically important crop in many tropical and subtropical countries and farmers in these countries may have challenges similar to the ones documented in this study. Broader multi-country studies would be important to confirm this. Economic gains from better knowledge dissemination, including training in integrated pest management, could be very substantial (Rahman et al., 2018). Future research could try to quantify these economic gains using an

experimental approach in which a sample of chili producers is randomly allocated to a treatment and control group. Such study would further strengthen the evidence basis for the importance of interventions promoting good agricultural practices in vegetable production.

5. Conclusion

This study assessed farmers' knowledge and perceptions of anthracnose disease in chili pepper and its management in Bangladesh. The results show that virtually all of the surveyed farmers encountered problems with anthracnose in chili pepper production and this reduced overall crop yield by about 4% for the average farmer. The findings point at the importance of capacity building among chili pepper producers in Bangladesh so that they are better able to identify the problem, better understand how it spreads and can apply appropriate management methods more effectively. These findings are likely to apply to other crops and other countries as well. Improving this knowledge through targeted extension programs can make a considerable positive impact on the livelihood of farmers producing chili pepper in low- and middle-income countries.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Abu Hayat Md Saiful Islam: Conceptualization, Supervision, Project administration, Data curation, Methodology, Visualization, Formal analysis, Writing - original draft, Writing - review & editing, Investigation. **Pepijn Schreinemachers:** Conceptualization, Methodology, Supervision, Writing - review & editing. **Sanjeet Kumar:** Funding acquisition, Project administration, Conceptualization, Writing - review & editing.

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